25901796 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR/IMPROVEMENT/REAPPEARANCE EXAMINATION APRIL 2026

SECOND SEMESTER

Multi-Disciplinary Courses (MDC) - MG2MDCSTA101 - DATA ANALYSIS USING JAMOVI AND INTRODUCTION TO R

(2024 ADMISSION ONWARDS)

Duration: 1 Hours **Maximum Marks: 35**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer all questions

Each question carries **2** marks

1. What is the formula for calculating the mean of a data set? [K] / [CO2]

   a). Sum of values / Number of values

   b). Sum of values x Number of values

   c). Sum of values + Number of values

   d). Sum of values - Number of values

2. Which type of data is measured on a scale where differences are meaningful but there is no true zero? [K] / [CO2]

   a). Nominal

   b). Ordinal

c). Interval

d). Ratio

3. What is the median of the following data set: 1,3,5,8,9? [U] / [CO2]

a). 5

b). 3

c). 9

d). 8

4. What is quartile deviation? [U] / [CO2]

a). A measure of central tendency

b). A measure of dispersion

c). A measure of skewness

d). A measure of kurtosis

5. What is the variance of the following data: 2,2,2,2,2? [U] / [CO2]

a). 2

b). 0

c). 10

d). 5

6. In logistic regression, which of the following is binary? [U] / [CO3]

a). Dependent variable

b). Independent variable

c). Error term

d). All of the above

7. Which of the following describes the relationship between variables when the correlation coefficient is +1? [K] / [CO3]

a). No relationship

b). Perfect negative relationship

c). Weak relationship

d). Perfect positive relationship

8. Which of the following can be included in an R Markdown document? [K] / [CO4]

a). R code

b). LaTeX

c). Text formatted with markdown

d). All of the above

9. Suppose we have two vectors: x= (10,20,30,40), y= (20,30,40,50).How can we create a data frame containing x and y? [U] / [CO4]

a). dataframe(x,y)

b). dataframe

c). data.frame(x,y)

d). create.data.frame(x,y)

10. Which one of the following is a comparison operator in R? [K] / [CO4]

a). >

b). +

c). *

d). /

[2x10 = 20]

Part B

Short Essay Type Questions

Answer any **3** questions

Each question carries **5** marks

11. Define standard deviation. Find standard deviation for the following dataset: 8,9,6,5,2. [A] / [CO2]

12. Define mode. Identify the mode for the following dataset:21,56,65,32,21,89,68,21,69,21. [A] / [CO2]

13. Distinguish between simple and multiple linear regression. [K] / [CO3]

14. Distinguish between comparison and assignment operators in R. [U] / [CO4]

15. Define any three built-in functions in R with examples. [K] / [CO4]

[5x3 = 15]